Pablo Arrighi

Senior Researcher in Quantum Computing — Scientific Advisor

+33 611257467
pablo.arrighi@universite-paris-saclay.fr
https://lmf.cnrs.fr/Perso/PabloArrighi

French-Uruguayan, speaks French, Spanish and English.

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Degrees

Univ. Grenoble Alpes/Habilitation à Diriger les Recherches
2009

Title: Quantum cellular automata.

University of Cambridge, U.-K./PhD
2000–2003

Title: Representations of quantum operations with applications to quantum cryptography

Imperial College, London U.-K./Bachelor of Engineering
1997–2000

First class degree in Information Systems Engineering.

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Permanent positions

Inria-Saclay/Research Director (1st class)
2024–

Research at the LMF laboratory.
Head of the QuaCS (Quantum Computation Structures) EPC Inria team.

Univ. Paris-Saclay/Full Professor (1st class)
2020–2024

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Evaluation & Advisory Experience

STAB of the Quanta Institute at AMU/Member of the committee
2026–

The Scientific and Training Advisory Board is composed of external scientific personalities and aims at providing the strategy of the Quanta Institute, which gathers the quantum activities of Aix-Marseille Université.

CoNRS Section 6/Member of the committee
2019–2021

The CoNRS, Section 6, is key to the scientific policy of CNRS in Informatics, e.g. selecting whom to recruit as a CNRS researcher, whom to promote, attributing the CNRS medals.

Quantum center of Saclay/Executive committee member

2020–

The center coordinates the French strategy for quantum technologies, in particular the QuantumEdu plan, at the scale of U. Paris-Saclay and Institut Polytechnique de Paris. Its ecosystem includes startups such as Quandela and Pasqal.

Bureau du Comité des Équipes Projet/Member

2024–

The BCEP evaluates the creation of new Inria and associated teams, as well as relationships with labs and doctoral schools.

Selection committee member

2025 Assistant Prof. position at AMU
2024 Permanent researcher positions Inria Lyon
2023 Chaire d'excellence – AMU
2023 Assistant Prof. position at Télécom Paris – IPP
2023 Permanent researcher positions of Inria-Alpes and Inria Lyon
2019 Aix-Marseille Univ. for ATER
2008, 2009 ENS-Lyon for ATER

GdR IQFA/Founding member, bureau member

2010–2014

The national research network structures the French community in quantum information across all disciplines, organizing yearly colloquiums and spreading key information.

HCERES expert, evaluating:

Dec. 2018 LTCI laboratory
March 2010 Strategy of Ecoles des Mines de Nantes
June 2008 Research strategy of the Univ. of Mulhouse

Referee for grant making bodies

2025 Slovenian ARIS, Swiss SNSF, Luxembourgish FNR
2024 Chilean Fondecyt, Luxembourgish FNR
2023 Belgian FNR
2016, 2017, 2018 Grand-Est region via European Science Foundation
2018 SIRTEQ
2017 Grenoble QuEng PhD program & Chairs of excellence
2017 EU CHIST-ERA projects (twice)
May. 2010, Jul. 2012, March 2019, 2021, 2022 Belgian FNRS projects and recruitment
2016 Italian PRIN-CINECA projects
2011, 2014 Uruguayan ANII projects
Dec. 2009 Digitéo projects

Other expertise

2024 "Shepherd" for the creation of the EPC Inria team Phiqus

2022 "Shepherd" for the creation of the EPC Inria team Quriosity

2022 RIPEC C3 expert for three applicants

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Industry Engagement

CIFRE PhD thesis with Quandela/Supervisor

2022–2025

Together with Benoît Valiron and Shane Mansfield as co-supervisors, PhD student Nicolas Heurtel is working on measurement-based quantum computing for photonics, developing a dedicated programming language with proper semantics, so that optimization can be done rigorously.

Atos-Fourier prize/Jury member

June 2018 See here.

DGA-REI contract/Leader

16 months, 2009–2010

This grant is now called ANR Astrid. I employed two software engineers full-time, at Floralis developing an implementation of a higher-order process algebra, aiming at a transfer of technology for the DGA.

Crédit Impôt Recherche/Expert

2006–2008

Evaluation of private sector research entitling to tax reductions, 3 companies.

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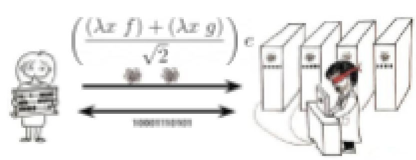
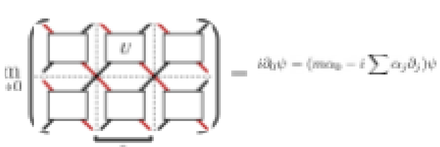
Some major contributions



Blind quantum computation. At a time when quantum cryptography research was focussing on the symmetric scenarios of secure-two-party computation and hitting the no-go result for unconditionally secure bit commitment, I was the first to shift the focus on the asymmetric scenario of "blind quantum computing". My protocol offers an efficient, quantum alternative to the later discovered classical homomorphic encryption scheme. The topic has become a full-blown research topic and enjoyed experimental demonstration.



Quantum walks for simulating particles. QW are discrete-time discrete-space unitary evolutions driving the propagation of a particle. They constitute numerical schemes to be run on quantum computers. We proved in full rigour, using numerical analysis methods, that some QW converge towards the dynamics of real-life particles, such as the propagation of the electron— thereby providing very simple, algorithmic explanations of their behaviours, as well as quantum simulation algorithms for them. The topic is a very active field of research, and we are leading the

	Terms:	$\mathbf{t},\mathbf{r},\mathbf{u} ::= \mathbf{b} \mid (\mathbf{t})\,\mathbf{r} \mid \mathbf{0} \mid \alpha.\mathbf{t} \mid \mathbf{t}+\mathbf{r}$	
	Basis terms:	$\mathbf{b} ::= x \mid \lambda x\,\mathbf{t}$	
Elementary rules:	*Factorisation rules:*	*Application rules:*	*Beta reduction:*
$\mathbf{t}+\mathbf{0} \to \mathbf{t}$,	$\alpha.\mathbf{t}+\beta.\mathbf{t} \to (\alpha+\beta).\mathbf{t}$	$(\mathbf{t}+\mathbf{r})\,\mathbf{u} \to (\mathbf{t})\,\mathbf{u}+(\mathbf{r})\,\mathbf{u}$	$(\lambda x\,\mathbf{t})\,\mathbf{b} \to \mathbf{t}[\mathbf{b}/x]$
$0.\mathbf{t} \to \mathbf{0}$,	$\alpha.\mathbf{t}+\mathbf{t} \to (\alpha+1).\mathbf{t}$	$(\mathbf{u})\,(\mathbf{t}+\mathbf{r}) \to (\mathbf{u})\,\mathbf{t}+(\mathbf{u})\,\mathbf{r}$	
$1.\mathbf{t} \to \mathbf{t}$,	$\mathbf{t}+\mathbf{t} \to (1+1).\mathbf{t}$	$(\alpha.\mathbf{t})\,\mathbf{r} \to \alpha.(\mathbf{t})\,\mathbf{r}$	
$\alpha.\mathbf{0} \to \mathbf{0}$,		$(\mathbf{r})\,(\alpha.\mathbf{t}) \to \alpha.(\mathbf{r})\,\mathbf{t}$	
$\alpha.(\beta.\mathbf{t}) \to (\alpha\times\beta).\mathbf{t}$,		$(\mathbf{0})\,\mathbf{t} \to \mathbf{0}$,	
$\alpha.(\mathbf{t}+\mathbf{r}) \to \alpha.\mathbf{t}+\alpha.\mathbf{r}$.		$(\mathbf{t})\,\mathbf{0} \to \mathbf{0}$.	

shift towards multi-particle quantum simulation.

Linear-algebraic λ-calculus and vectorial type systems. Whilst partly unexplained, it is clear that the algorithmic speed-up of Quantum Computation arises by tapping into the parallelism granted by the superposition principle. The idea of a vector space of computations arises quite naturally in this context. We designed the Linear-algebraic λ-calculus, merging higher-order computation, be it terminating or not, in its simplest and most general form (namely the untyped λ-calculus) with linear algebra in its simplest and most general form also (the oriented axioms of vector spaces we obtained beforehand), and proved its confluence. The language allows for quantum control structures, and superpositions in the order of application of functions. We devised vectorial type systems, where the term $\alpha.t+\beta.u$ has type $\alpha.T+\beta.U$. The topic has become a full-blown research topic.

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Scientific production

42 international-level refereed journals including *J. of Computer and Systems Sciences, Int. J. Found. of Computer Science, Mathematical Structures in Computer Science, Information and Computation, Logical Methods in Computer Science, Quantum Information Processing, New Journal of Physics, Quantum, Physical Review Letters*.

31 international-level refereed conferences including *ICALP, QIP, MFCS*.

13 invited talks at international-level conferences.

3 book chapters.
Cf. up-to-date list.

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Scientific recognition

Prizes

2024 Ranked 1st at the admissibility of the Inria Research Director external recruitment competition
2021 First class professor (PR1) on the national contingent (CNU 27)
2009–2017, 2020-2024 PEDR
2016 Prime d'Engagement pédagogique for this course.
2000–2003 Cambridge European Trust fellowship
2000–2003 Cambridgecourse Isaac Newton Trust fellowship
2002 European Union Marie Curie fellowship
2000 Psion Prize for the best final year project, on Automatic signature extraction for mimetic viruses

Editor

2023– Theoretical Computer Science

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Scientific diffusion

Popular science, see an up-to-date list.

4 articles in *Physics World*, *Cosmos*, *La Recherche* (*2*), *Programmez*.

3 book chapters e.g. in *Current Trends in Theoretical Computer Science*

5 columns on my work in *New Scientist*, *Pour la Science*, MIT Technology review and APS Physics.

1 interview *France Info*.

1 podcast RDV Tech.

2 round-table discussion at industrial congress *Systematic* and at the Argentinian Ministry of Science.

2 popular science talks e.g. for CS high school teachers or at *La commanderie*.

6 Philosophy of science seminars including at *ENS Ulm; La Sorbonne; Cerisy-la-Salle.*